Binomial and Normal (Solutions)

Specific types of probability distributions

How should we model our population?



If we have full information,
we can define a specific
random variable
(e.g. Apgar score).

Otherwise, we make assumptions about the population and choose a pre-defined random variable “template”.

A **binomial** is a type of discrete RV that models a sample from a binary population.

A **normal** is a type of continuous RV with a bell-curve shape.

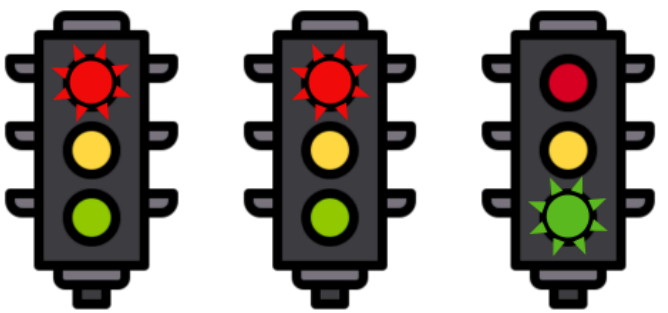
Both models give us formulas for statistical analyses.

A binomial is used to model certain binary populations.

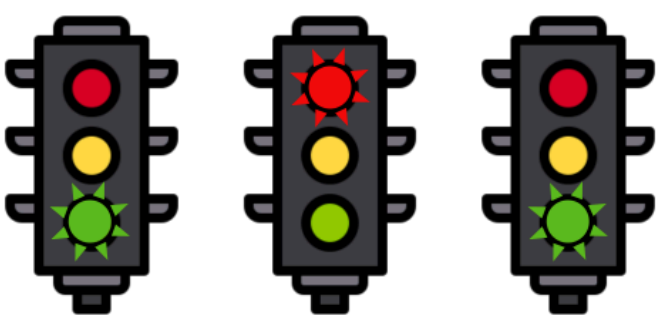
When driving home, we go through 3 stoplights, which are either red or green.



If X is the number of green lights we get out of three, what is the distribution of X?



x = 1



x = 2

To actually come up with a distribution, we need some simplifying assumptions. Assume:

- Lights are independent
- Each one has probability 0.6 of being green

Exercises

Outcomes	x	$P(X = x)$
RRR	0	(0.4)(0.4)(0.4) = 0.064
	1	
GGR, GRG, RGG	2	3(0.6)(0.6)(0.4) = 0.432
	3	

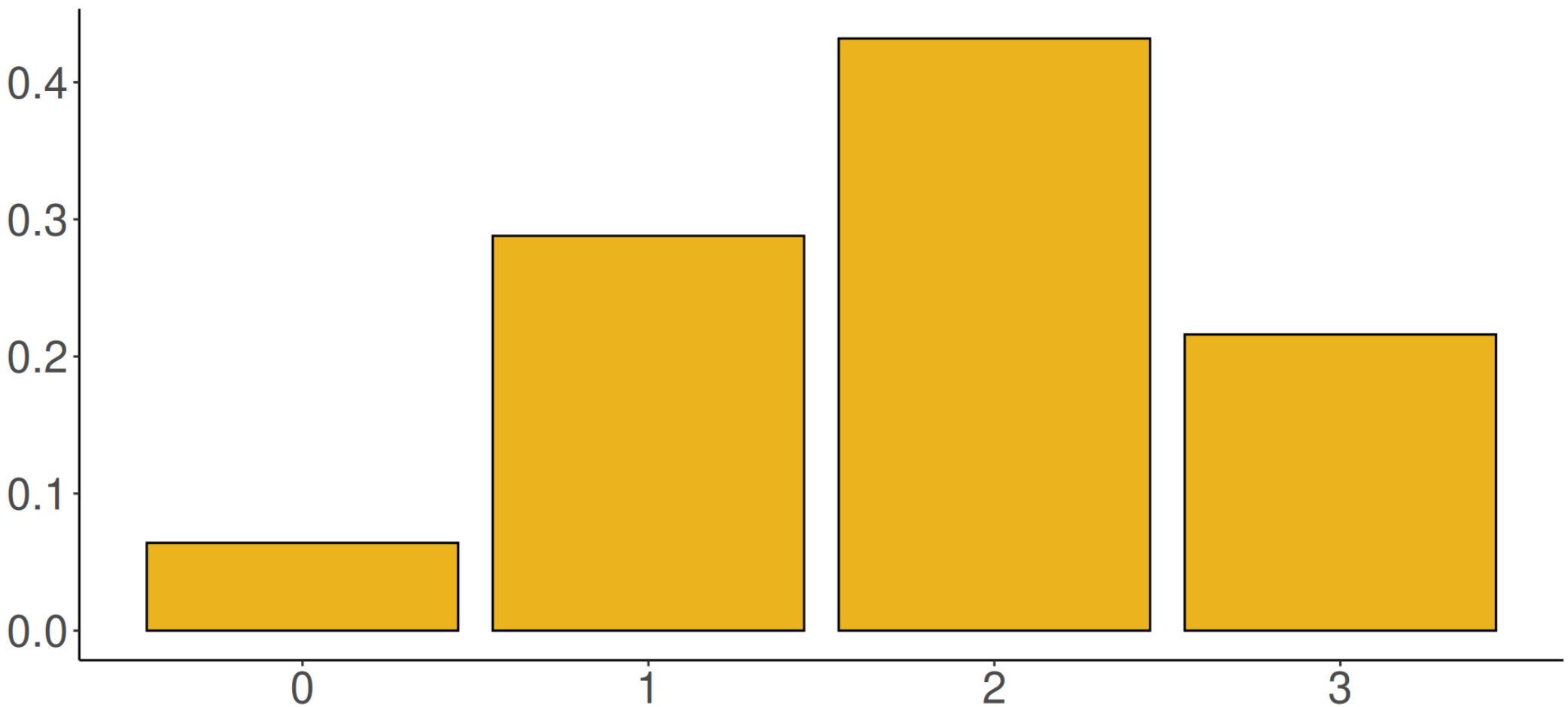
- Complete the pmf of X, the number of green lights out of 3. Make sure it adds up to 1.



Solutions/Hints

- $X = 1$ occurs with outcomes GRR, RGR, and RRG, and has probability 3(0.6)(0.4)(0.4) = 0.288.
- $X = 3$ occurs with outcomes GGG, and has probability (0.6)(0.6)(0.6) = 0.216.

X = Green lights out of 3



X is a **binomial** random variable, written $X \sim Binom(3, 0.6)$.

In general, a binomial RV counts the number of "successful" draws from a sample from a binary population.

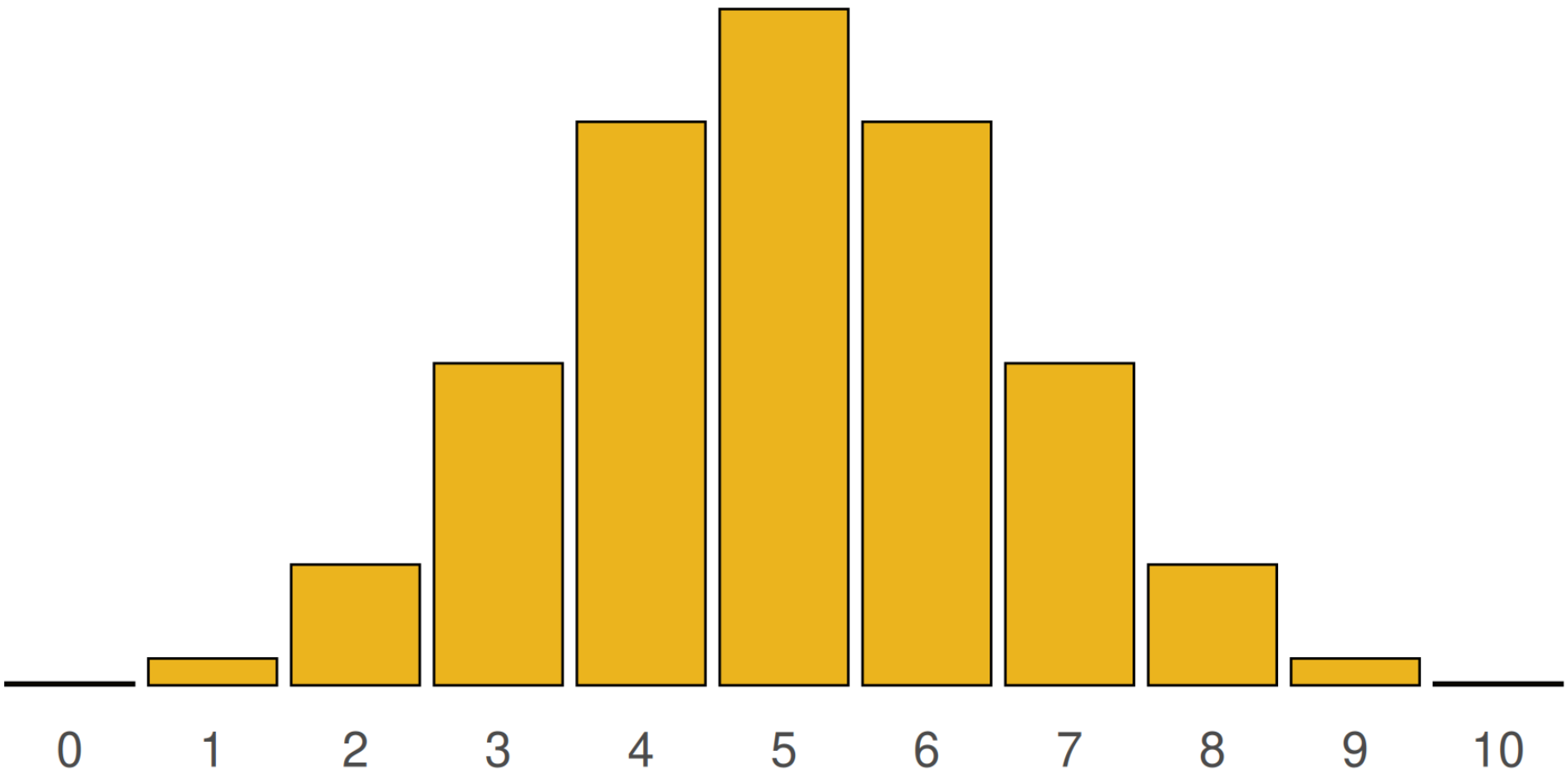
$X \sim Binom(n, p)$

When is it valid to use a binomial model? We had to make some assumptions about the traffic lights.

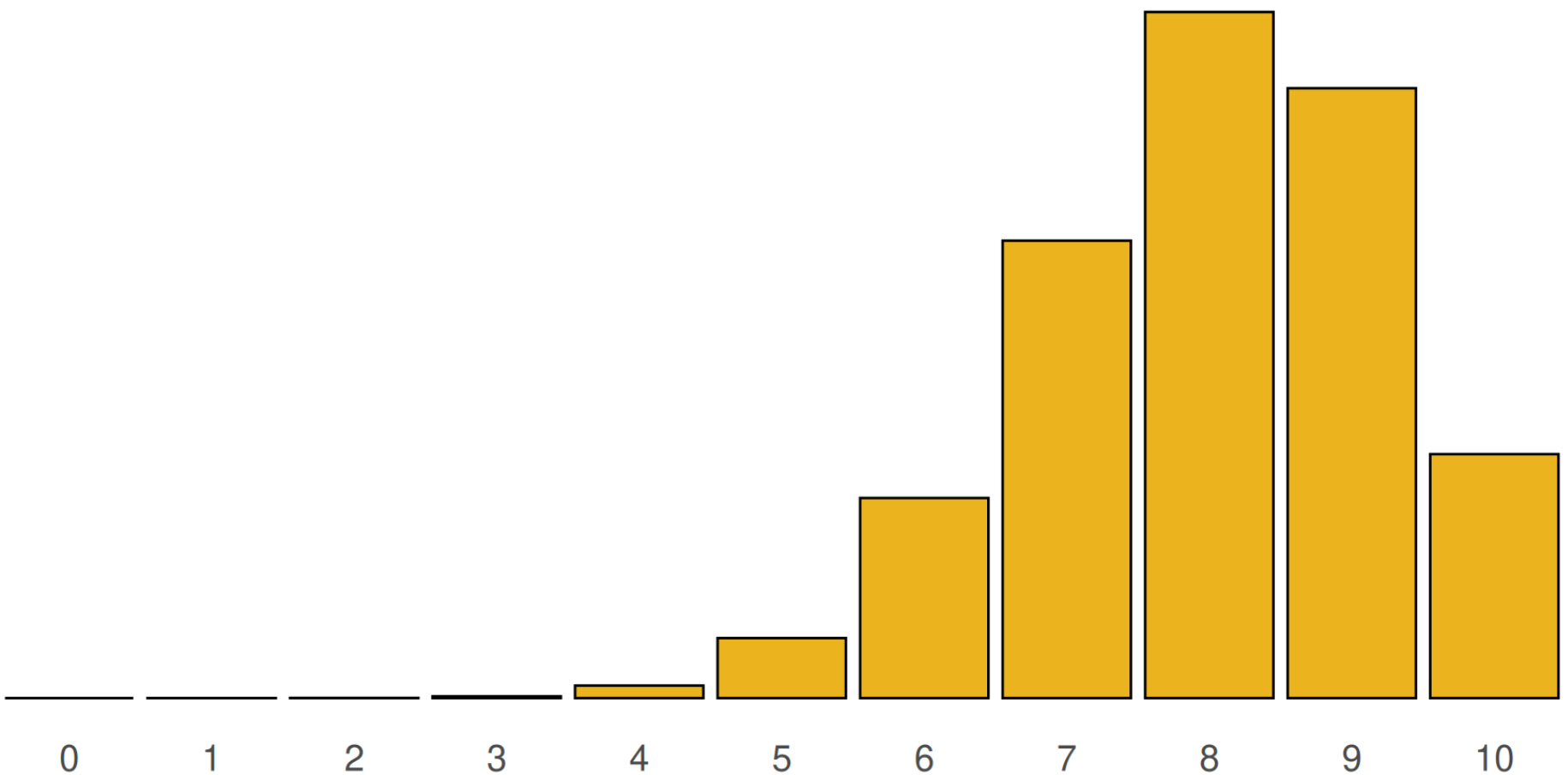
Formally, the draws:

- **B**: are binary (success, or failure)
- **I**: are independent
- **N**: have a fixed sample size n
- **S**: have the same probability p

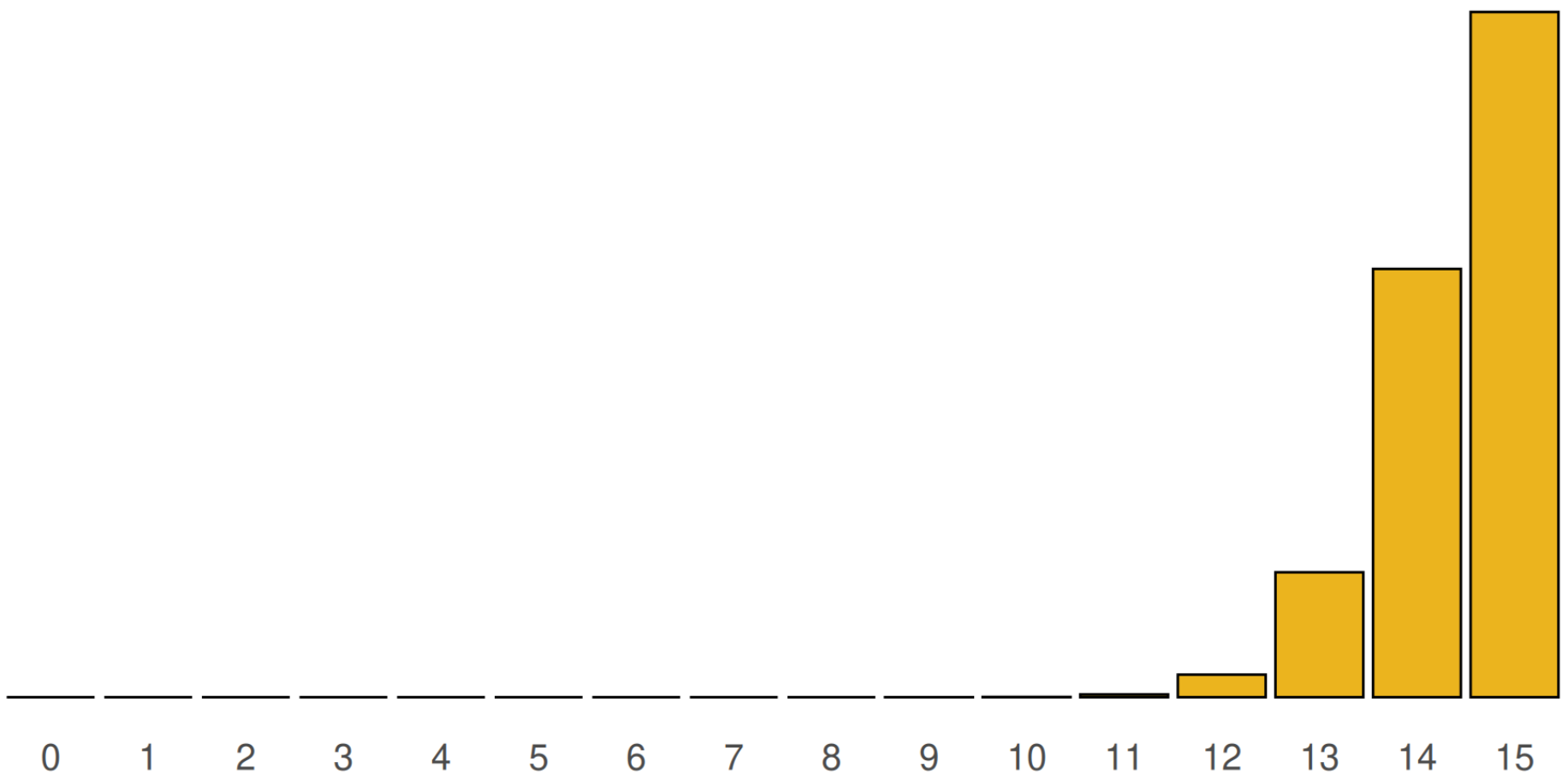
$Binom(n = 10,\ p = 0.5)$



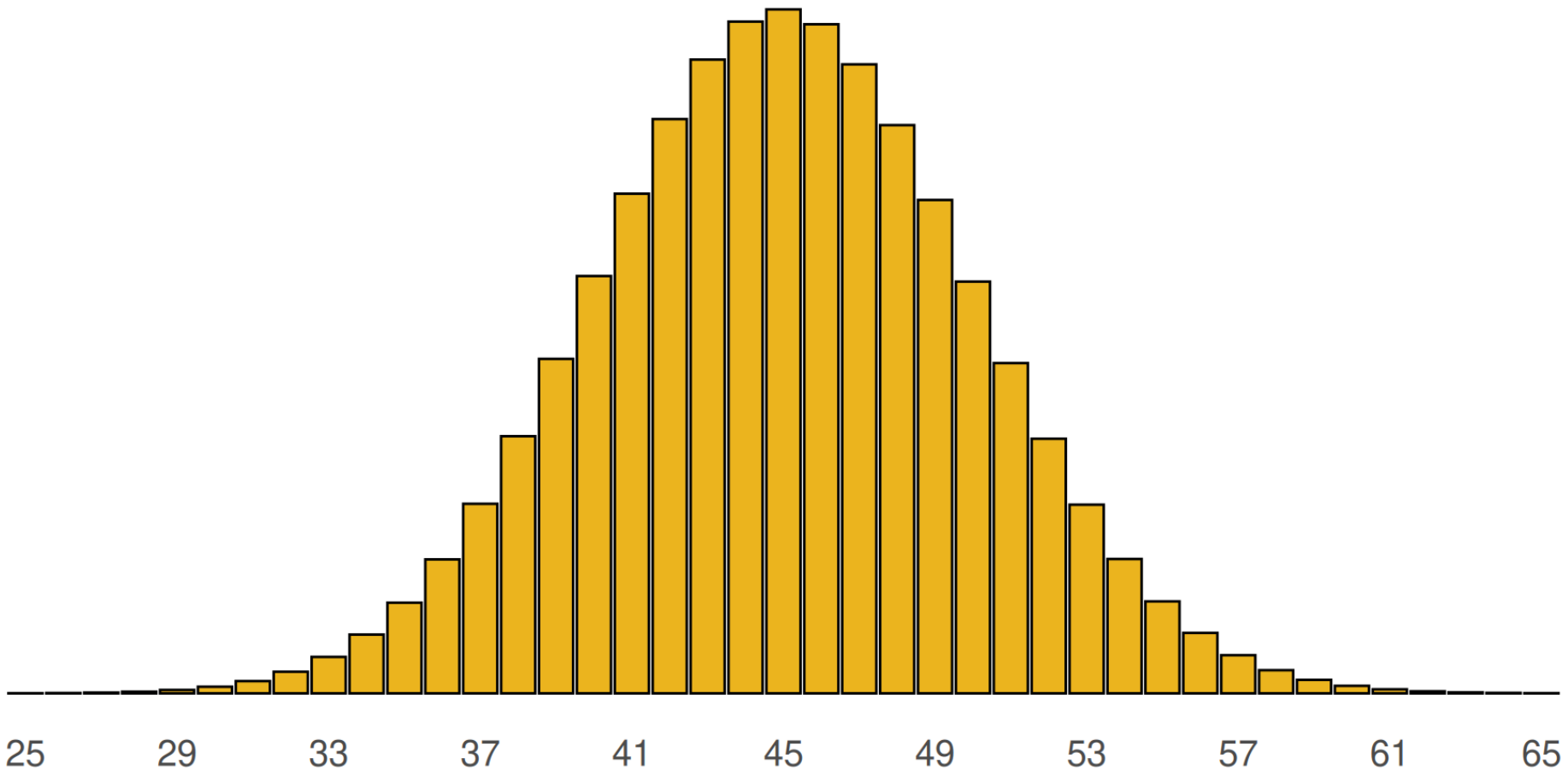
$$Binom(n = 10,\ p = 0.8)$$



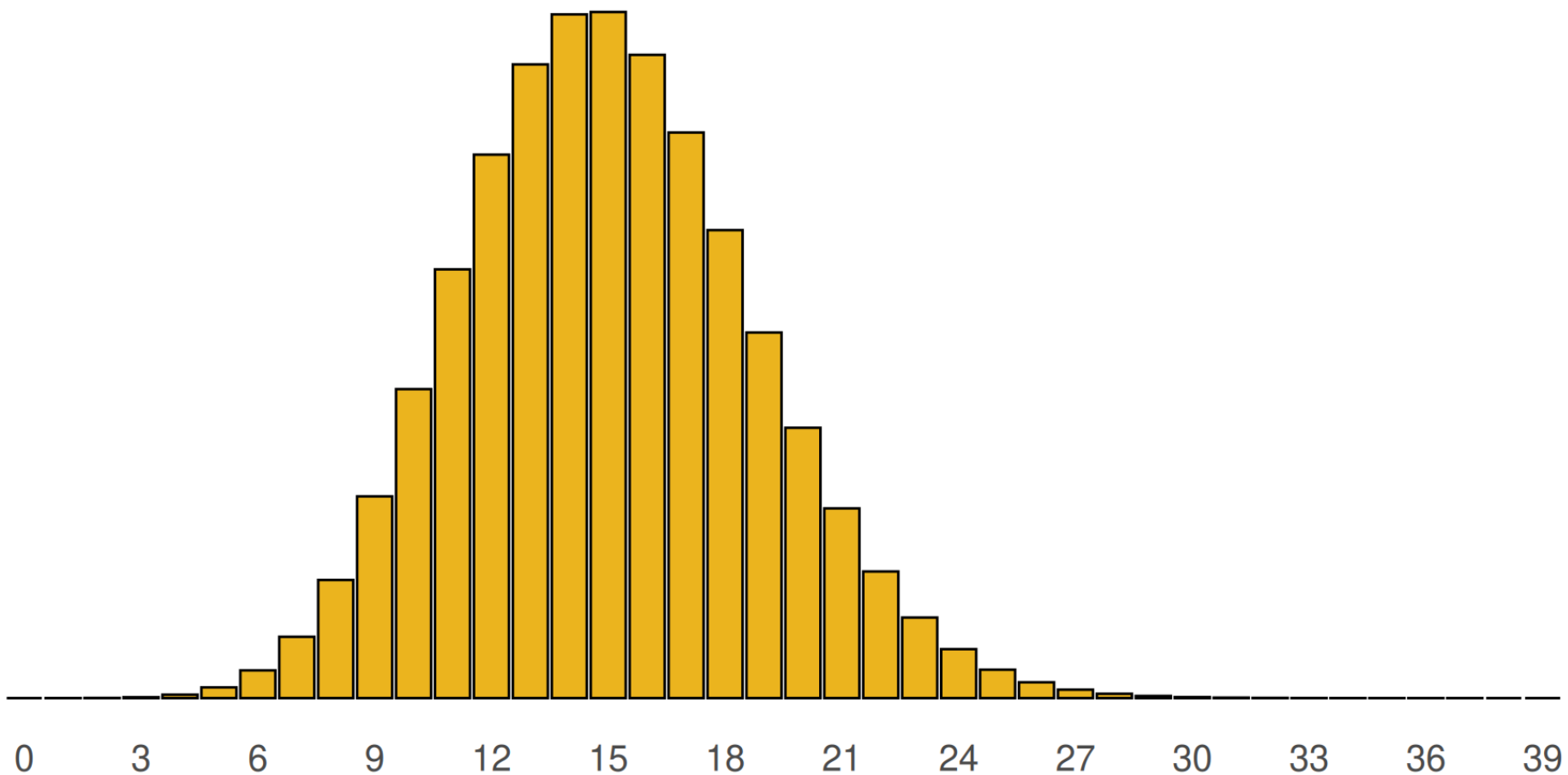
$$Binom(n = 15,\ p = 0.96)$$



$$Binom(n = 100,\ p = 0.45)$$



$$Binom(n = 300,\ p = 0.05)$$



We have seen something like this before! In the powerplant example, we took a sample of 300 births and had hypotheses

$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p \neq 0.5$$

How would this look if we used a binomial model for simulation?

In this case, we don’t actually have to use simulation at all!

We have a formula for the binomial pmf. In the traffic lights example,

$$P(X = 2) \; = \; 3(0.6)^2(0.4)^1$$

3 ways, 2 successes, 1 failure

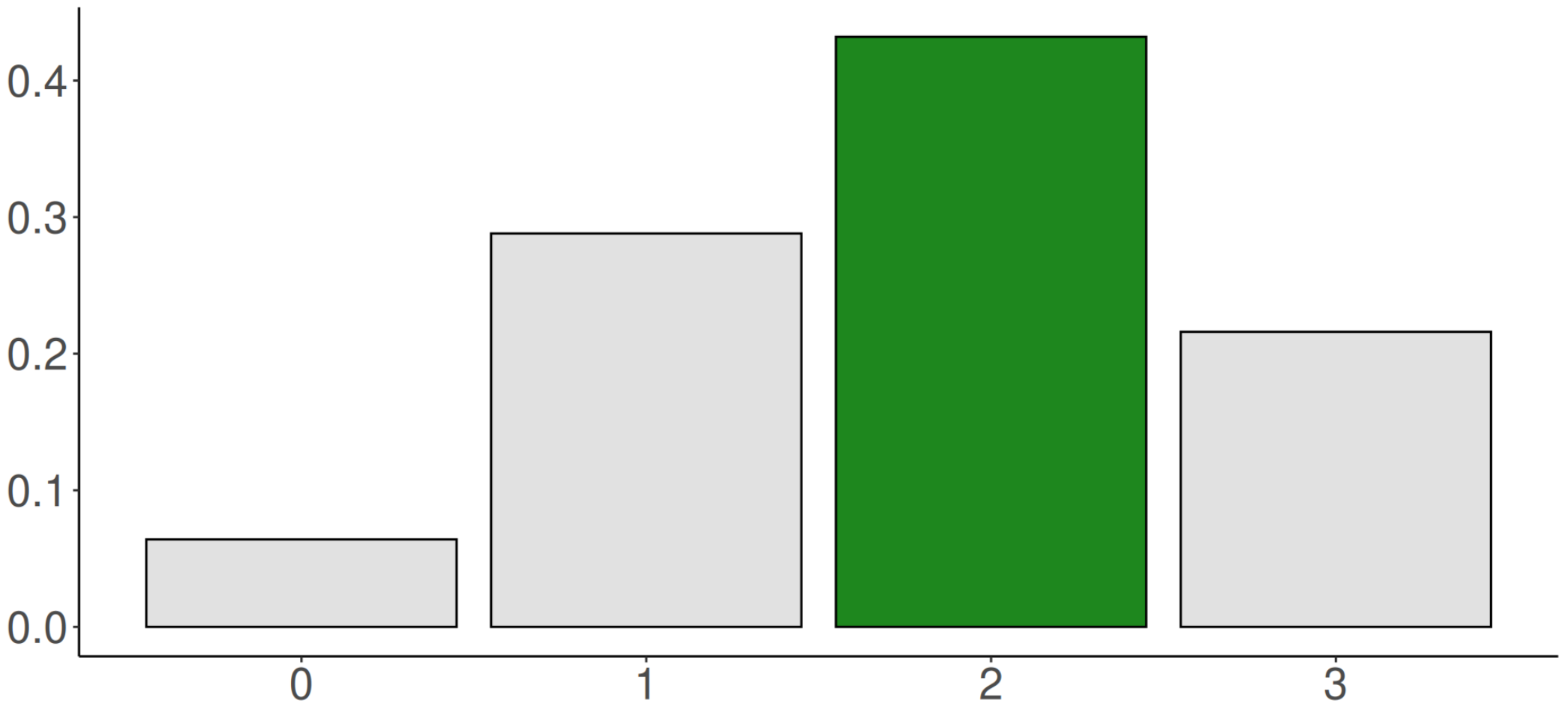
In general, for $X \sim Binom(n, p)$:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

“n choose x” ways, x successes, n-x failures

In R, we use `dbinom()`.

```
dbinom(2, size = 3, prob = 0.6)
```

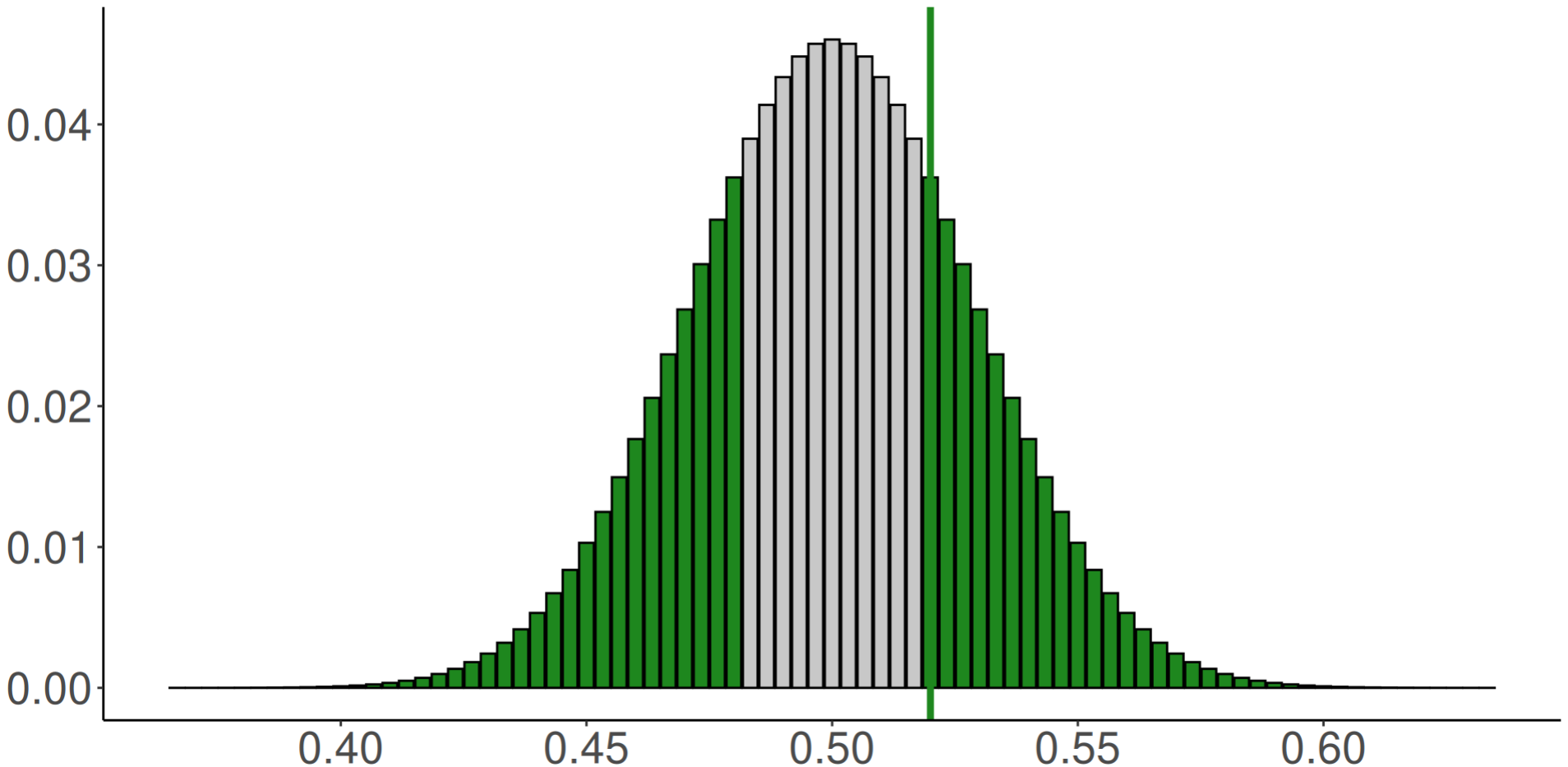


$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p \neq 0.5$$

In the power plant example, we observed 52%, or 156 girl births. The other interesting outcome is 48%, or 144 girl births.

We can add up `dbinom()` values to calculate a p-value without simulation.

```
sum(dbinom(0:144, 300, 0.5)) +
sum(dbinom(156:300, 300, 0.5))
```



Exercises

A psychic claims to be able to guess the suit of a card without knowing. In 200 card draws, they get 57 correct. If p is their true probability of getting the suit right, test hypotheses

$$H_0 : p = 0.25 \quad \text{versus} \quad H_A : p \neq 0.25$$

at the 5% level.

- First, simulate a null distribution with `replicate()` and find the p-value.
- Next, re-calculate the p-value with `dbinom()`. Are the two methods consistent?
- CHALLENGE: What if we wanted to build a 99% confidence interval for p based on 57/200? How could we find M_{lower} and M_{upper} with the exact binomial distribution?



Solutions/Hints

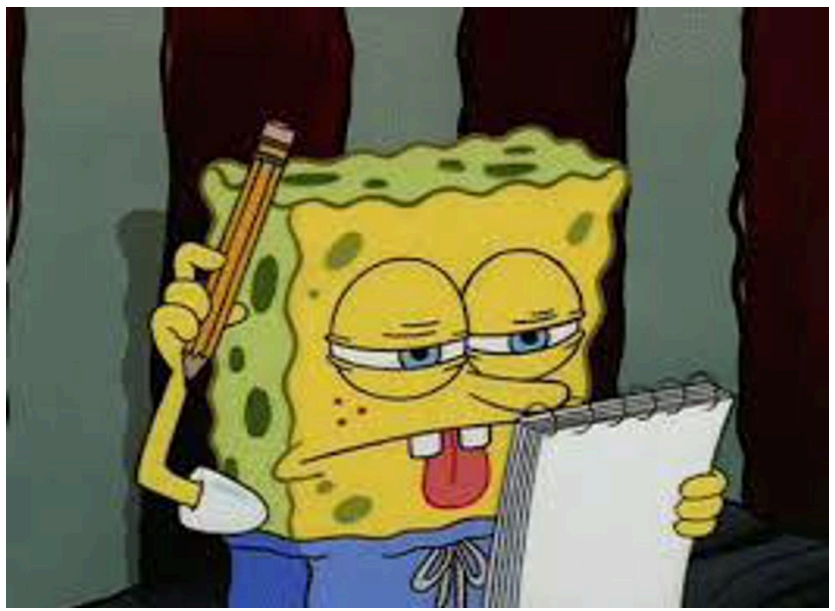
- The simulation example is something we've seen before - replicate 10000 samples with a 25% success rate, then find the probabilities outside of 43/200 and 57/200 for a two-sided p-value.
- To find the p-value from the exact binomial distribution, find the probability that Binom(200, 0.25) is ≤ 43 or ≥ 57.
- If we wanted to build a 99% CI based on the exact distribution Binom(200, 57/200), we would need to find the 0.5 and 99.5 percentiles of that discrete binomial variable.

p, the unknown true proportion, "links" the binomial model to our data. This is the **parameter**.

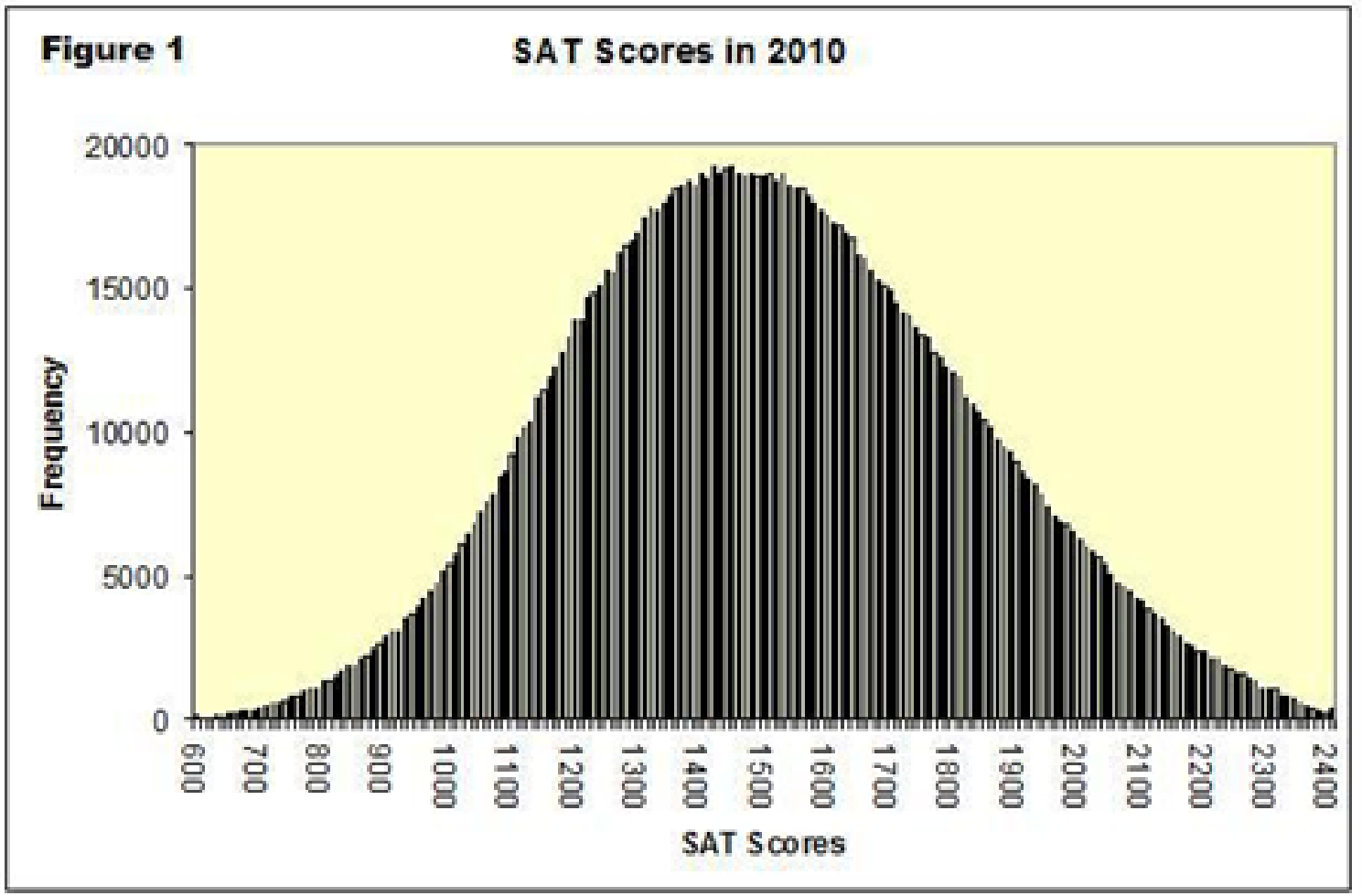
Parameters define our model. For example, a binomial model has parameters n and p.

The "parameter of interest" is the unknown population value we are trying to study.

Another very popular model in statistics is a **normal random variable**. This is a bell-curve shaped continuous distribution.

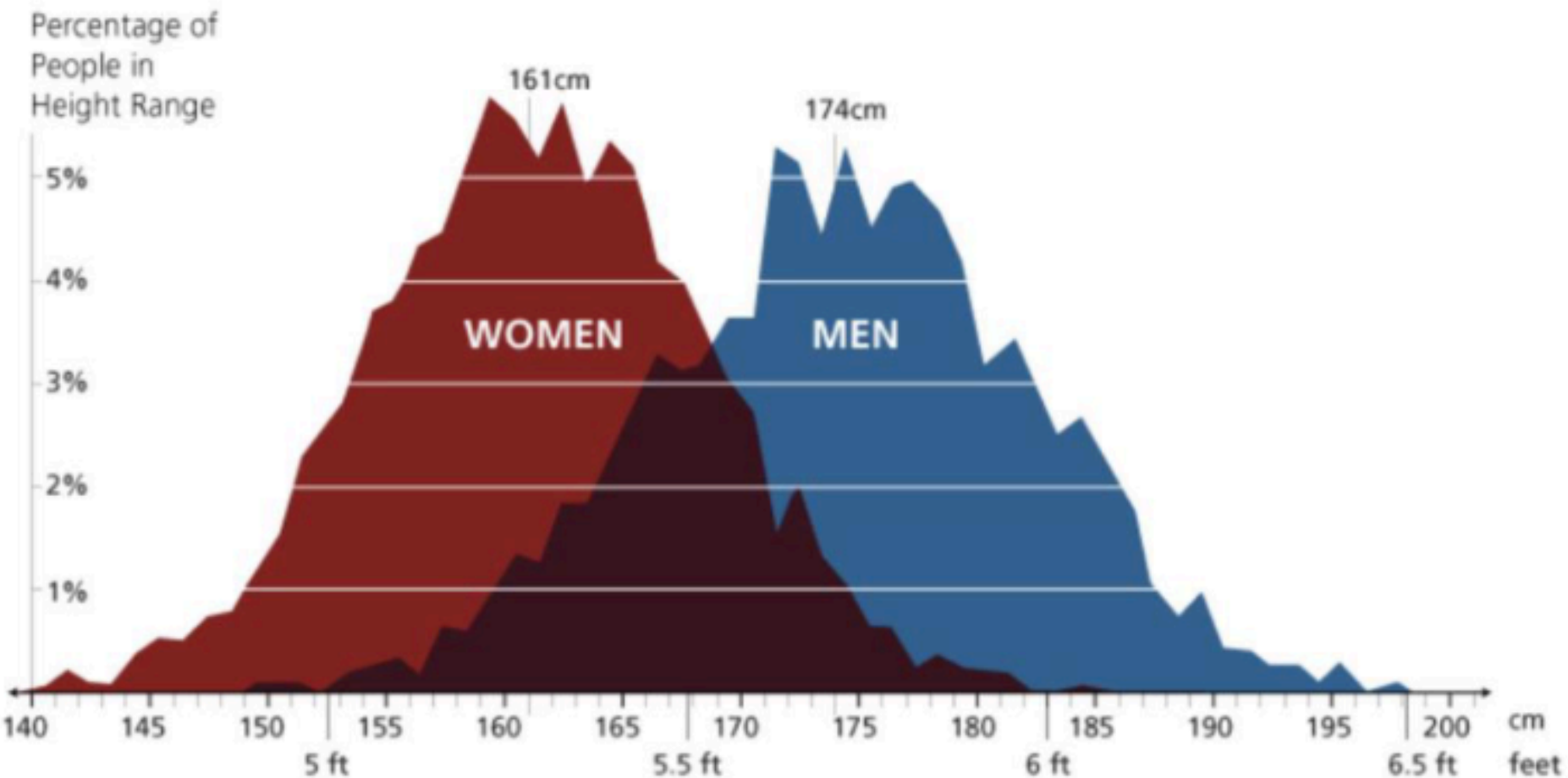


Consider SAT scores. Most students score somewhere in the middle, with a few people scoring much less or much more.

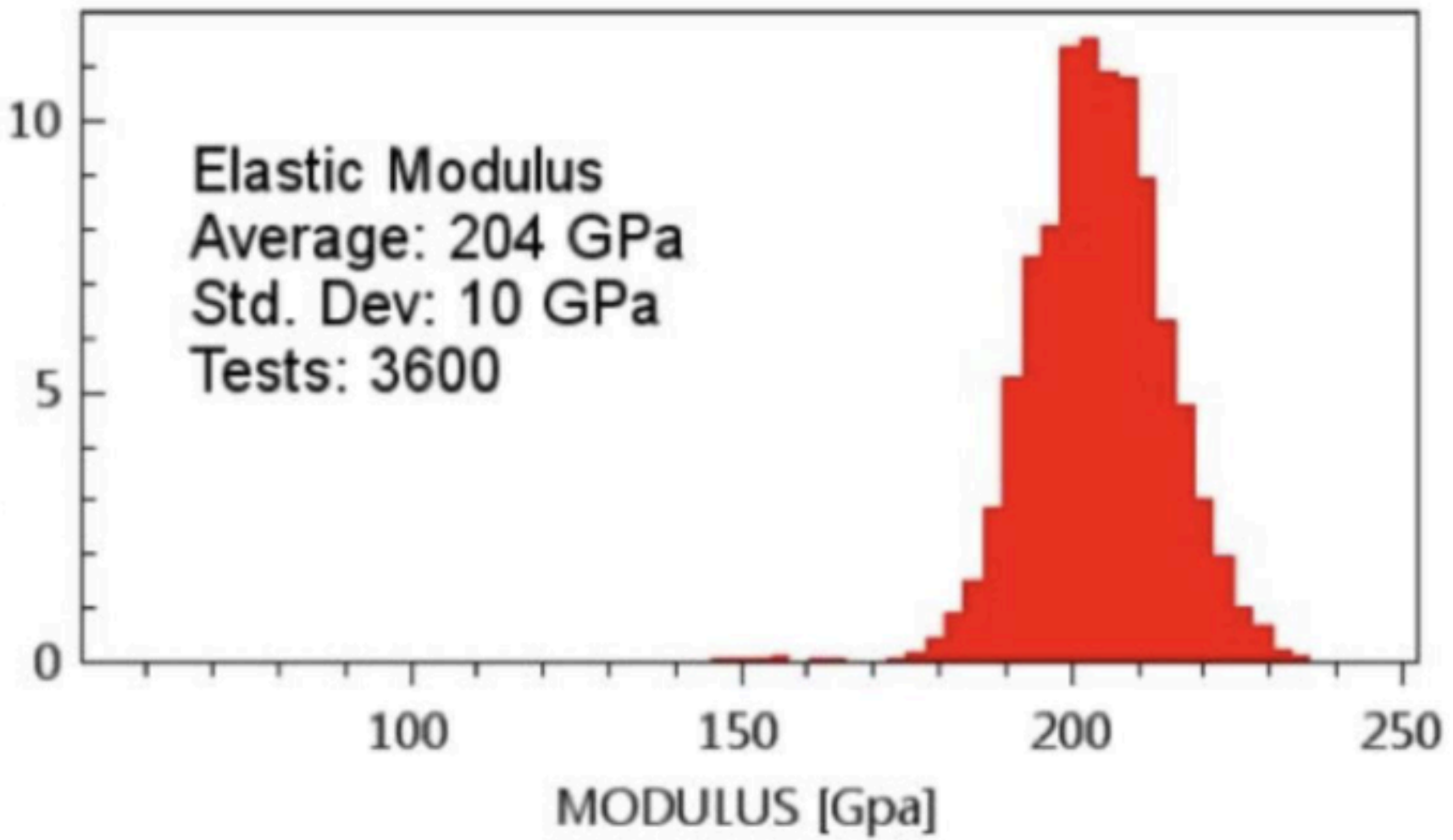


Where else does this bell-curve shape come up?

Heights/body measurements:



Engineering:



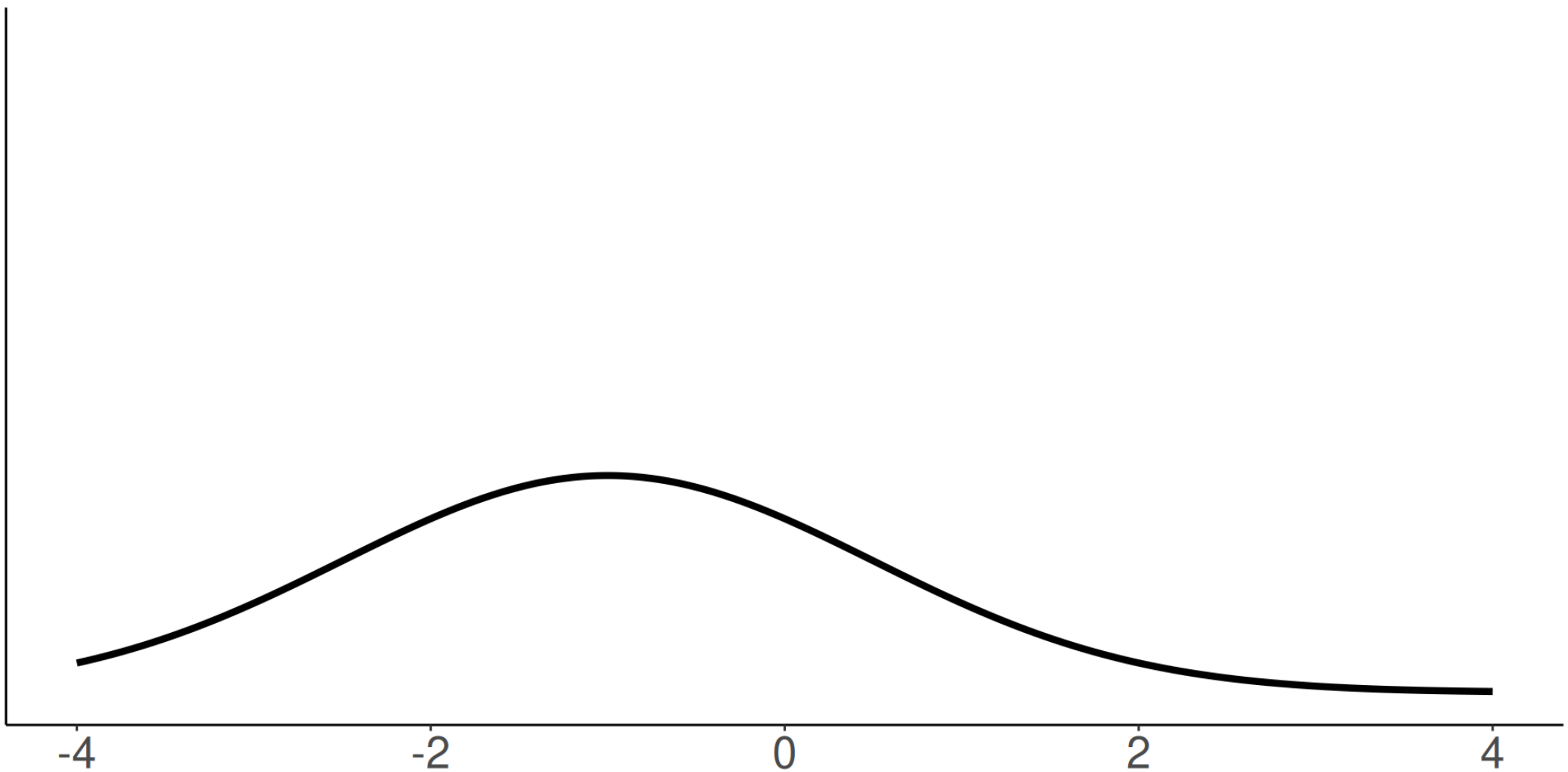
All normal RVs have the same underlying shape, just a different center (μ) and spread (σ).

Write $X \sim N(\text{mean},\ \text{sd})$ or $X \sim N(\mu, \sigma)$.

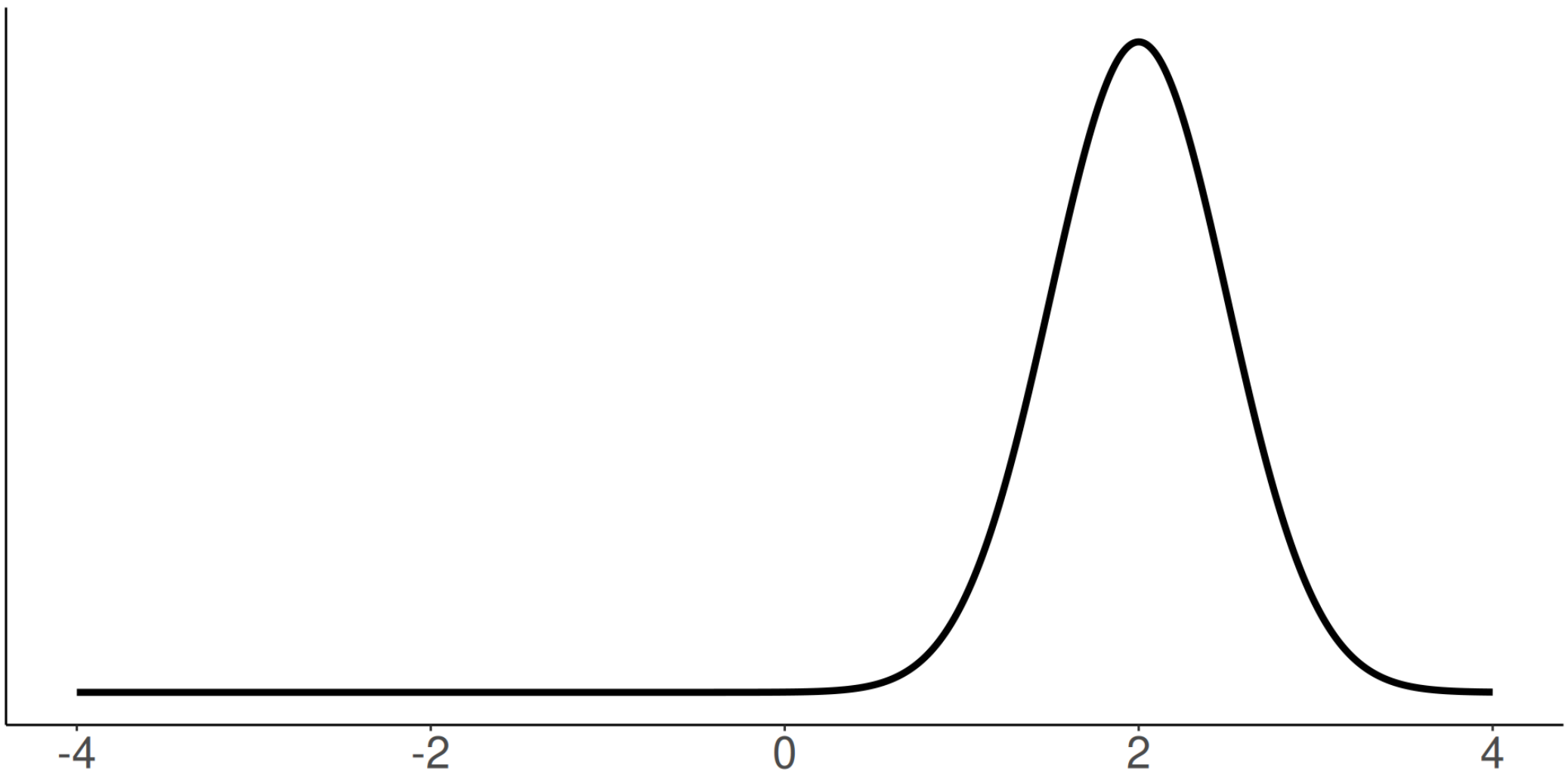
$N(0, 1)$



$N(-1, 1.5)$



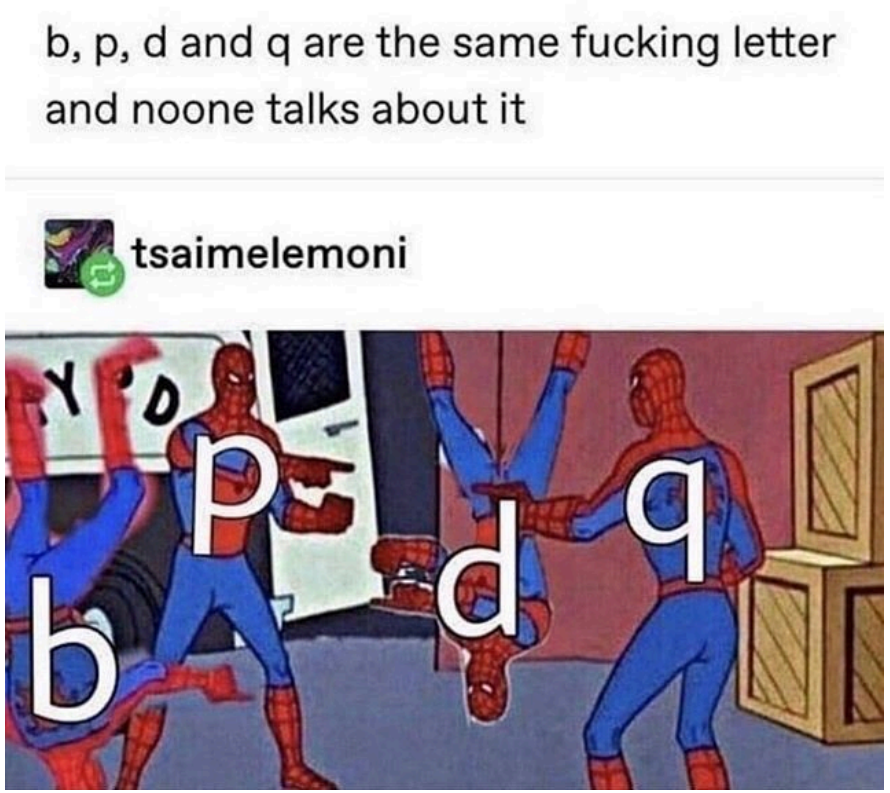
$N(2, 0.5)$



The bell curve (Normal pdf) is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

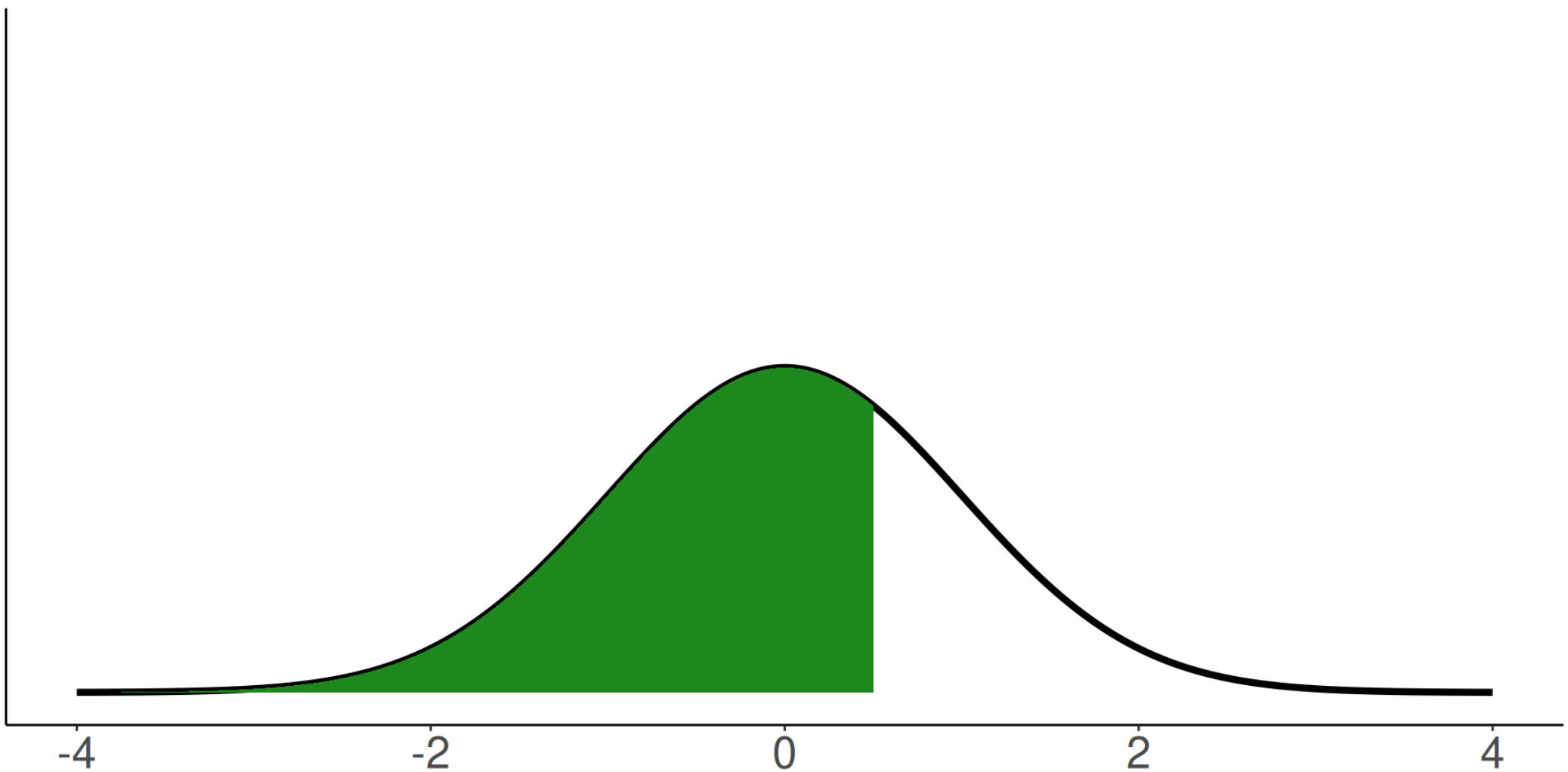
- Parameterized by μ, σ
- Goes to $(-\infty, \infty)$
- Has a smooth bell-curve shape



Probabilities are given by the R function `pnorm()`.

Quantiles are givne by `qnorm()`.

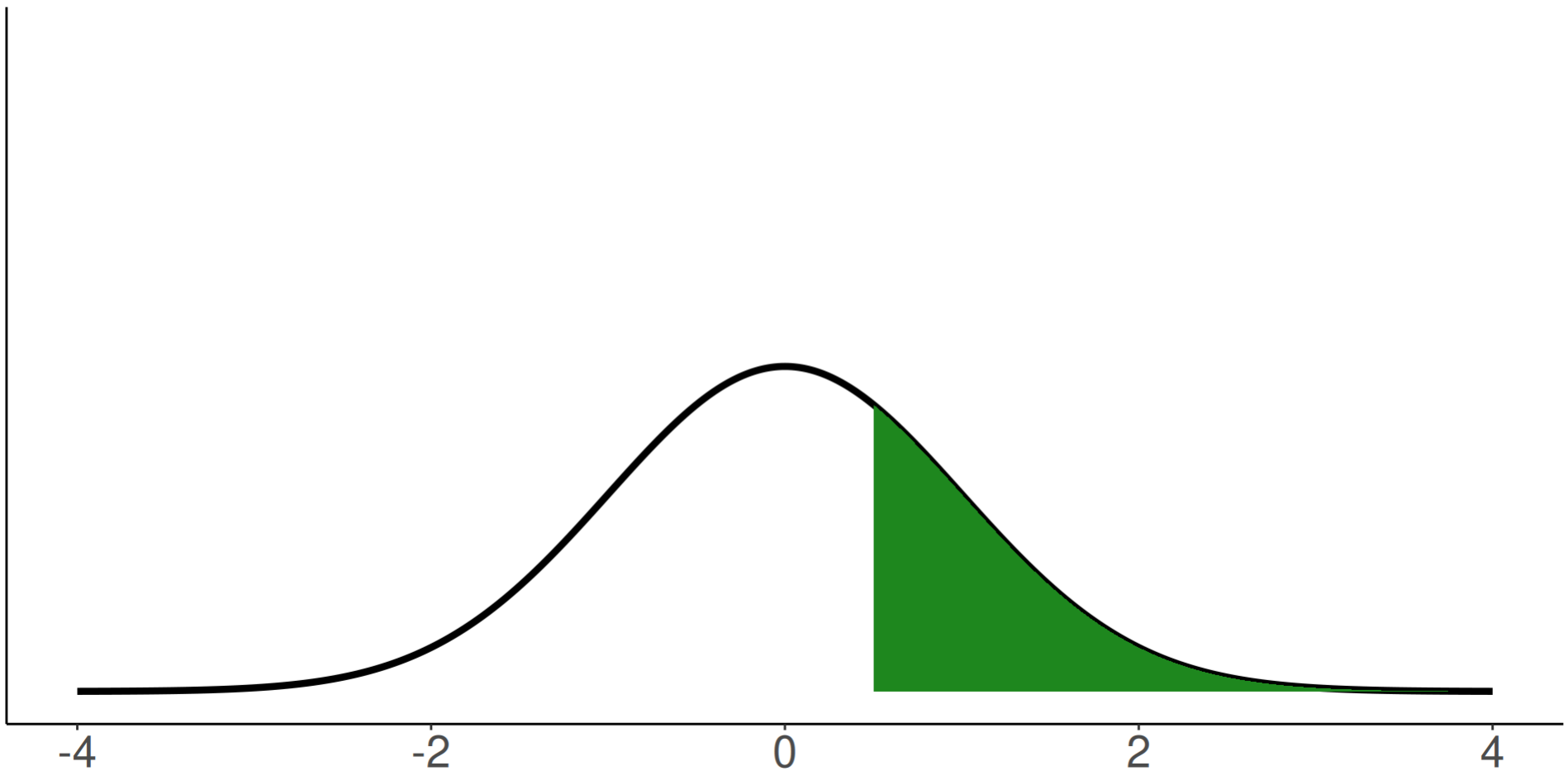
```
pnorm(0.5, 0, 1)
```



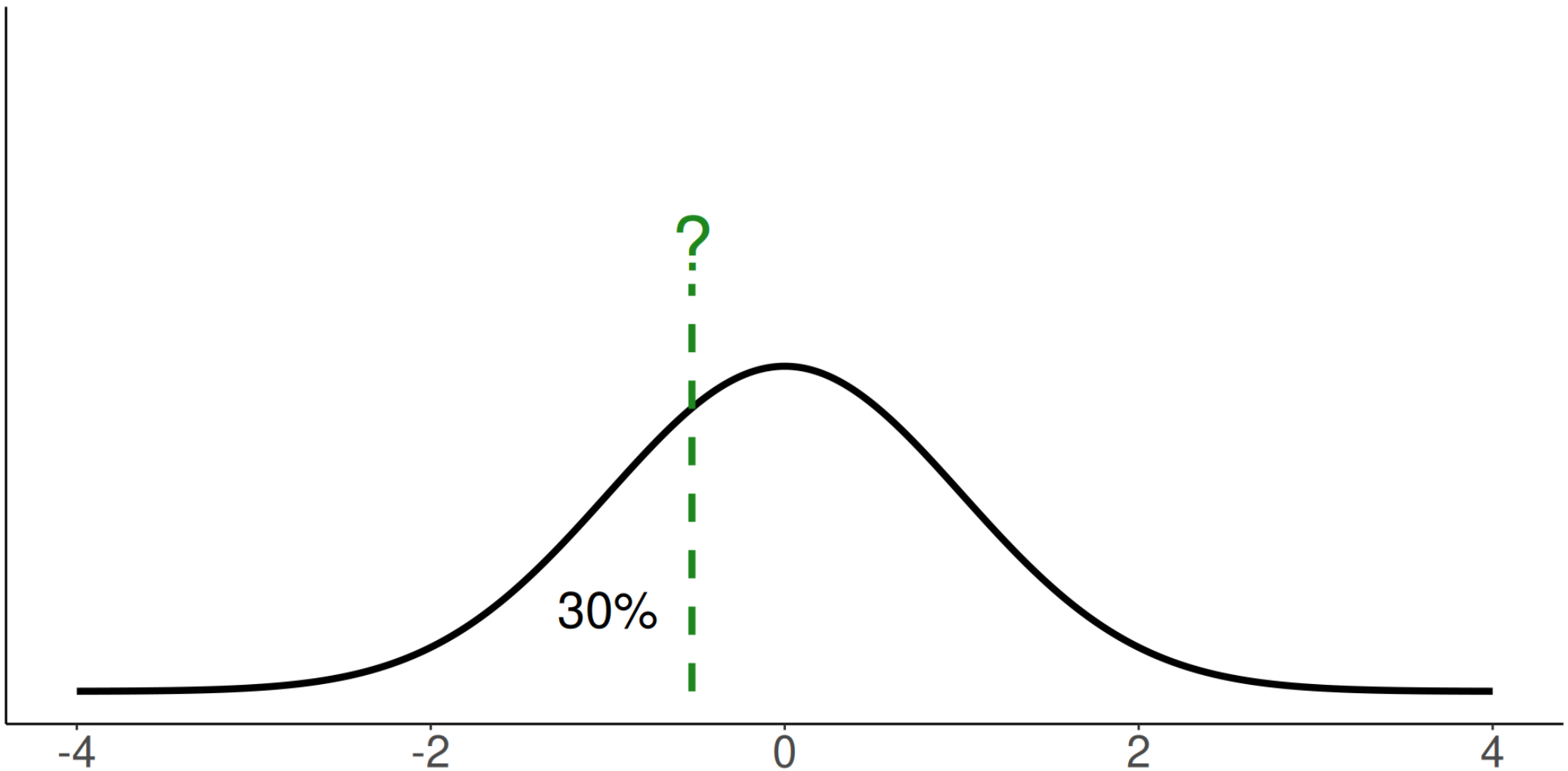
```
1 - pnorm(0.5, 0, 1)
```

or

```
pnorm(0.5, 0, 1, lower.tail = FALSE)
```



```
qnorm(0.3, 0, 1)
```



Consider two normal RVs: $X_1 \sim N(100, 25)$, $X_2 \sim N(10, 7)$. Which is more likely?

- $P(X_1 > 125)$
- $P(X_2 > 24)$

All normal RVs have the same underlying shape.

Standard deviations, **z-scores** are a “universal language” for normal RVs.

A z-score is a standardized value. For $X \sim N(\mu, \sigma)$,

$$P(X \leq x) \;=\; P(Z \leq \frac{x-\mu}{\sigma}) \;=\; P(Z \leq z)$$

where $Z \sim N(0,1)$.

$Z \sim N(0,1) \approx$ “how many SDs away from the mean”?

Exercises

The weight of flour in a batch of dough is $F \sim N(500, 12)$.
The weight of water in a batch of dough is $W \sim N(350, 4)$.

- A flour weight of 476 corresponds to what weight of water?
- Try to answer this question two ways: both on paper and with R.



Solutions/Hints

- You should find a weight of water that has a standard score of $\frac{476-500}{2} = -2$.
- You can also use `pnorm(476, 500, 12) %>% qnorm(350, 4)`.

The normal distribution arises naturally when studying certain sampling distributions.

Return to the D&D example, where we plotted the average of six ability scores.

Values naturally “converge” to the center.

More generally, let $X \sim D(\mu, \sigma)$ be any population.

The **Central Limit Theorem** says that the sample mean $\bar{X}$ will be normal for large enough n:

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The CLT also works for binomials! (A binomial is a type of sum)

If $B \sim Binom(n, p)$,

$$B \dot{\sim} N\left(np, \sqrt{np(1-p)}\right)$$

as long as $np(1-p) \geq 10$.

Exercises

Consider the central limit theorem. For large enough n, the distribution of $\bar{X}$ is normal.

- Why does the normal approximation work better as n gets larger?
- If the CLT applies to the sample mean, why does it also apply to the sample total?
- Why does the CLT not apply to the median or other measures?



Solutions/Hints

- As we take a larger sample, the sample mean averages out to be very close to the true mean. This effect does not happen for the median, which is discrete.
- The sample total is just the sample mean multiplied by a constant value n. So if the mean is a bell curve, the total is also a bell curve.

Up next: Statistical methods based on the normal distribution.

These methods assume that the population has a “nice” shape. They also require us to study the mean and SD specifically.

Are these things true?

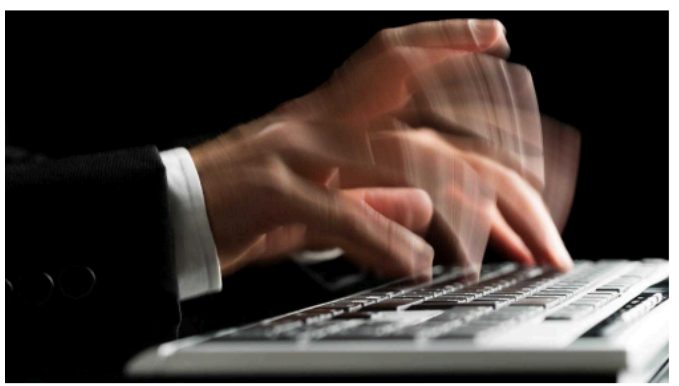
In general, to perform a hypothesis test:

1. Write hypotheses
2. Find a null distribution
3. Calculate p-value from our data

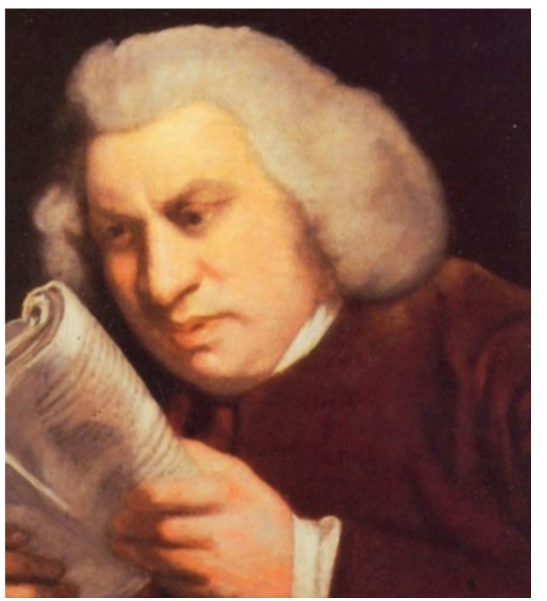
2. Find a null distribution

Simulation can generate a null distribution of any shape.

now



back then



Classical, formula-based methods assume the null is normal (or related).

Looking ahead

In chapter 10, we'll learn about using the normal distribution to perform inference on a single proportion. If you want a head start, consider a binomial RV

$$X \sim Binom(n, p)$$

that represents a sample from a binary population. Now, consider the sample proportion $\hat{p} = \frac{X}{n}$ which is a binomial divided by n. What is the mean and standard deviation of $\hat{p}$? If X is approximately normal, is $\hat{p}$ approximately normal too?